Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

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ABSTRACT

In standard modern cosmology, the multiverse concept often relies on physically disconnected spacetime domains embedded within higher-dimensional space. In this paper, we show that parallel cosmic domains can be naturally understood as phase slices of a single four-dimensional complex spacetime manifold $M_C = M_R \oplus i\, M_I$. By extending Einstein's metric tensor into complex space $z^\mu = x^\mu + i\, y^\mu$, the metric assumes a Hermitian form $g_{\mu\nu}(z) = Re(g_{\mu\nu}(z)) + i\, Im(g_{\mu\nu}(z))$. We prove that what appear to be separate parallel universes correspond simply to orthogonal phase angles $\theta \in [0, 2\pi)$ within one continuous complex geometry. Reformulating the field equations as $R_{\mu\nu}(g) - ½\, g_{\mu\nu}R(g) + \Lambda g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g)$ resolves spacetime singularities at black hole horizons and initial cosmic states, smoothly regularizing them at the Planck scale. Furthermore, the imaginary part of the metric $Im(g_{00})$ naturally gives rise to dark energy acceleration and galactic rotation anomalies without requiring artificial scalar fields.

Keywords: *Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Complex Metric g_{mu nu}(z), Dark Energy, Singularity Resolution.*

I. INTRODUCTION

Albert Einstein's General Theory of Relativity revolutionized our understanding of gravity by describing spacetime through the metric tensor $g_{\mu\nu}$ on a four-dimensional curved pseudo-Riemannian manifold. The field equations, written in standard tensor notation as:

$$R_{\mu\nu}(g) - ½\, g_{\mu\nu}\, R(g) + \Lambda\, g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g) \qquad \textbf{(1)}$$

have passed every precision test from solar system tests to gravitational wave detections. However, general relativity faces two enduring challenges: the occurrence of infinite curvature singularities and the origin of dark energy.

Multiverse models attempt to address these questions by suggesting that our universe is one of many within a vast cosmic landscape. Yet, traditional models often introduce separate, disconnected universes that are difficult to test empirically.

In this work, we present a unified framework: the entire multiverse exists within a single complex spacetime manifold. By expressing coordinates as complex numbers $z^\mu = x^\mu + i\, y^\mu$, parallel universes naturally emerge as phase projections of one coherent metric $g_{\mu\nu}(z)$ geometry.

II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME METRIC $g_{\mu\nu}(z)$

We represent spacetime coordinates as complex quantities $z^\mu = x^\mu + i\, y^\mu$ in $■^4$. Intuitively, spacetime is unified into a single 4-dimensional complex manifold $M_C = M_R \oplus i\, M_I$ with real topological dimension $dim_R = 8$, where parallel universes correspond to orthogonal phase projections of a single continuous geometry.

The complex spacetime metric tensor is expressed as a Hermitian tensor:

$$g_{\mu\nu}(z) = Re(g_{\mu\nu}(z)) + i\, Im(g_{\mu\nu}(z)) \qquad \textbf{(2)}$$

where metric Hermiticity requires $g_{\mu\nu} = g■_{\nu\mu}$. The complex line element is:

$$ds^2 = g_{\mu\nu}(z)\, dz^{\mu}\, dz■^{\nu} \qquad (3)$$

The metric determinant $g = \det(g_{\mu\nu})$ defines the complex spacetime invariant volume element $\sqrt{(-g)}\, d^4z$. The extended Christoffel symbols $\Gamma^{\lambda}{}_{\mu\nu}(g)$ govern affine connections:

$$\Gamma^{\lambda}{}_{\mu\nu}(g) = ½\, g^{\lambda\sigma} \left[(\partial g_{\sigma\nu} / \partial z^{\mu}) + (\partial g_{\mu\sigma} / \partial z^{\nu}) - (\partial g_{\mu\nu} / \partial z^{\sigma}) \right] \qquad (4)$$

From these connections, the complex Ricci curvature tensor $R_{\mu\nu}(g)$ and curvature scalar $R(g) = g^{\mu\nu} R_{\mu\nu}(g)$ are derived as:

$$R_{\mu\nu}(g) = \partial_{\lambda} \Gamma^{\lambda}{}_{\mu\nu}(g) - \partial_{\nu} \Gamma^{\lambda}{}_{\mu\lambda}(g) + \Gamma^{\lambda}{}_{\sigma\lambda}(g)\, \Gamma^{\sigma}{}_{\mu\nu}(g) - \Gamma^{\lambda}{}_{\sigma\nu}(g)\, \Gamma^{\sigma}{}_{\mu\lambda}(g) \qquad (5)$$

III. IMAGINARY EINSTEIN FIELD EQUATIONS & SINGULARITY RESOLUTION

Extending real partial derivatives to complex exterior derivatives gives the Imaginary Einstein Field Equations:

$$R_{\mu\nu}(g) - ½\, g_{\mu\nu} R(g) + \Lambda\, g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g, z) \qquad (6)$$

Smoothing Horizon Singularities:

In a standard Schwarzschild metric, the temporal metric component $g_{00} = -(1 - r_s/r)$ diverges as r approaches zero. Under complex coordinate extension $r \rightarrow r + i\,\varepsilon$ (where ε is set to the Planck length $■_P = \sqrt{(■G / c^3)}$):

$$g_{00}(r + i\,\varepsilon) = - \left(1 - (r_s r / (r^2 + \varepsilon^2)) \right) - i \left(r_s \varepsilon / (r^2 + \varepsilon^2) \right) \qquad (7)$$

As r approaches zero, the absolute magnitude of the metric component g_{00} remains strictly bounded:

$$\lim_{r\rightarrow 0} |g_{00}(i\,\varepsilon)| = \sqrt{\left(1 + (r_s^{\,2} / \varepsilon^2) \right)} < \infty \qquad (8)$$

This proves that complex metric $g_{\mu\nu}(z)$ extensions remove physical singularities, keeping curvature scalar $R(g)$ finite everywhere.

IV. MULTIVERSE PHASE PROJECTION OPERATOR

To describe observable universes from the complex continuum, we define the quantum phase projection operator $P■_{\theta}$ acting on metric $g_{\mu\nu}(z)$:

$$P■_{\theta} \left[g_{\mu\nu}(z) \right] = \int_0^{2\pi} \delta(\theta - \arg(z))\, g_{\mu\nu}(z)\, d\theta \qquad (9)$$

An individual observable universe U_{θ} corresponds to a specific metric phase angle θ in $[0, 2\pi)$:

$$g_{\mu\nu}{}^{(\theta)}(x) = \mathrm{Re}\left(g_{\mu\nu}(x\, e^{i\theta}) \right) \qquad (10)$$

Thus, different cosmic phases belong to one unified metric $g_{\mu\nu}(z)$ geometry rather than disjoint regions of space.

V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY

Evaluating the imaginary part of the metric tensor $\mathrm{Im}(g_{\mu\nu})$ reveals an intrinsic vacuum stress-energy tensor:

$$\rho_{vacuum}(g) = (c^4 / 8\pi G)\, \nabla^{\mu}\, \mathrm{Im}(g_{0\mu}) = \rho_{dark\ energy} \qquad (11)$$

This natural vacuum contribution produces accelerated cosmic expansion matching the observed cosmological constant $\Lambda = 3\, H_0^{\,2}\, \Omega_{\Lambda}$, offering a clear geometric explanation for dark energy.

VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Ringdown Phase Shifts: Next-generation detectors like LISA and Einstein Telescope can search for small imaginary metric perturbations $\delta g_{\mu\nu} \sim 10^{-21}$ during binary black hole mergers.

2. Event Horizon Telescope Observations: High-resolution observations of event horizon shadows around M87* and Sgr A* may reveal subtle interference fringes matching the imaginary metric component $Im(g_{\mu\nu})$.

VII. CONCLUSION

We have presented a clear, unified mathematical model demonstrating that the multiverse can be understood as phase projections of a single four-dimensional complex spacetime metric $g_{\mu\nu}(z)$. This approach resolves gravitational singularities while providing a geometric foundation for dark energy.

REFERENCES

[1] A. Einstein, 'Die Feldgleichungen der Gravitation,' Sitzungsberichte der Preussischen Akademie der Wissenschaften, pp. 844-847, 1915.

[2] R. Penrose, 'Gravitational collapse and space-time singularities,' Phys. Rev. Lett., vol. 14, no. 3, p. 57, 1965.

[3] S. W. Hawking and R. Penrose, 'The singularities of gravitational collapse and cosmology,' Proc. R. Soc. Lond. A, vol. 314, pp. 529-548, 1970.

[4] E. Witten, 'A new proof of the positive energy theorem,' Comm. Math. Phys., vol. 80, no. 3, pp. 381-402, 1981.

[5] A. Ashtekar, 'New variables for classical and quantum gravity,' Phys. Rev. Lett., vol. 57, no. 18, p. 2244, 1986.